INCH-POUND
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DETAIL SPECIFICATION

Plug, Shock Attenuating and Plug, Lifting, for M107 and M804

This specification is approved for use by the U.S. Army Armament Research Development and Engineering Center (ARDEC), and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements for the manufacturing of Shock Attenuating Lifting Plugs and Lifting Plugs for the M107 and M804 (see 6.1).

1.2 Classification. The plugs are described as follows:

1.2.1 Plug, shock attenuating. The shock attenuating lifting plug can be used on any 155MM HE projectile.

1.2.2 Plug, lifting, for M107 and M804. The M107 and M804 lifting plug is to be used only on the M107 and M804 155MM projectiles because it does not provide the required protection other projectiles require.

Comments, suggestions, or questions on this document should be addressed to the Commander, U.S. Army ARDEC, ATTN: AMSRD-AAR-QES-E, Picatinny, NJ 07806-5000 or emailed to ardec-stdzn@pica.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to assure the completeness of this list, document users are cautioned that they must meet all requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

TT-C-490 – Chemical Conversion Coatings and Pretreatments for Ferrous Surfaces (Base for Organic Coatings)

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1168 – Ammunition Lot Numbering and Ammunition Data Cards

MIL-STD-1916 – DoD Preferred Methods for Acceptance of Product

(Unless otherwise indicated, Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC) DRAWINGS

9341742 – Plug, Shock Attenuating
13015329 – Plug, Lifting, M107

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(Copies of these drawings may be requested online at Drawing-Request@pica.army.mil or from U.S. Army ARDEC, ATTN: AMSRD-AAR-AIS-TD, Picatinny, NJ 07806-5000.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article inspection. When specified (see 6.2) a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on drawing 9341742 or 13015329.

3.2.1 Plug, Shock Attenuating. The materials, components and assemblies shall be in accordance with dwg. 9341742.

3.2.2 Plug, Lifting for M107 and M804. The materials, components and assemblies shall be in accordance with dwg. 13015329.

3.3 Ammunition lot numbering. Ammunition lot numbers shall be assigned in accordance with MIL-STD-1168.

3.4 Tension. The plug shall sustain a load of 8000 pounds minimum without any sign of failure.

3.5 Ductility. The Plug shall withstand permanent deformation of the eye without cracking.

3.6 Fracture. The plug shall be free from porosity and shrinks or discontinuous structure.

3.7 Workmanship. All parts shall be fabricated and finished in a thorough, workmanlike manner. They shall be clean and free of burrs, sharp edges, unblended radii, cracks, surface defects, chips, dirt, grease, dust and other foreign matter. Where cleaning and preparation are required by component details, the finish specified shall apply to the indicated surfaces regardless of contour configuration. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All paint shall be dry to the touch before packing for shipment. The paint shall be considered dry when the soft tacky condition no longer exists and the paint feels firm when touched with a finger using moderate pressure.

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4. VERIFICATION

TABLE I. Requirement/Verification Cross Reference Matrix.

Methods of Verification N/A – Not Applicable 1 – Analysis 2 – Demonstration 3 – Examination 4 – Test							Classes of Verification A- First Article B- Acceptance		
Section 3 Requirements	Description	Verification Method					Verification Class		Section 4 Verification
		N/A	1	2	3	4	A	B	
3.1	First Article Test				X	X	X	X	4.2
3.2	Components and assemblies				X	X	X	X	4.3
3.2.1	Plug, shock attenuating				X	X	X	X	4.3.2.1/4.4.2
3.2.2	Plug, Lifting, for M107 and M804				X	X	X	X	4.3.2.2/4.4.2
3.3	Ammunition lot numbering				X		X	X	4.3.1
3.4	Tension					X	X	X	4.3.2.1/4.4.1 4.4.1.1
3.5	Ductility					X	X	X	4.3.2.1/4.4.1 4.4.1.2
3.6	Fracture					X	X	X	4.3.2.1/4.4.1 4.4.1.3
3.7	Workmanship				X		X	X	4.3.2.1/4.3.2.2

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4.1 Classification of verification. The verification requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection.

4.2.1 First article quantity. First article verification shall be performed on the quantity of items as indicated in table II.

4.2.1.1 Multiple cavity molds. When a multiple cavity mold is to be used in production, the sample shall include at least three plugs from each cavity in the mold. The plug shall exhibit permanent cavity identification.

4.2.2 Inspections to be performed. The first article verification shall be performed in accordance with Table II.

4.2.3 First article rejection. If any item fails to comply with requirements, the first article sample shall be rejected.

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TABLE II. FIRST ARTICLE INSPECTION

EXAMINATION OR TEST	NUMBER OF SAMPLES	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Plug, Shock Attenuating			
Dwg. 9341742			
Examination for defects	25 min, Note 1	3.2.1	4.3.2.1
Salt spray test	2	3.2.1	4.4.2
Plug, Lifting, for M107 and M804			
Dwg. 13015329			
Examination for defects	25 min, Note 1	3.2.2	4.3.2.2
Salt spray test	2	3.2.2	4.4.2

1. Minimum of 3 per cavity from multiple cavity molds.

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4.3 Conformance inspection.

4.3.1 Lot formation. Lot formation shall be in accordance with lot formation requirement of MIL-STD-1916 (Formation and identification of lots or batches).

4.3.2 Classification of characteristics.

a. Sampling requirements. Inspection sampling requirements for critical, major and minor characteristics are defined in MIL-STD-1916. Unless specified otherwise, Inspection Level IV shall be used for all characteristics defined as Majors and Inspection Level II for all Minor characteristics; Critical characteristics shall be addressed in accordance with MIL-STD-1916.

b. Conformance inspection. Conformance inspection shall be performed in accordance with paragraph 4.3.2.1 and 4.3.2.2. For all conformance inspections the same sample specimen may be used for all non-destructive examinations or tests unless otherwise specified.

c. Alternative Inspections. Alternative conformance inspections may be submitted and approved in accordance with MIL-STD-1916.

d. Tool control. Dimensions marked tool control shall be gaged at the beginning of production and whenever tooling is replaced. Where destruction of components is necessary to inspect these dimensions, measurements of the tool may be substituted provided there is an established correlation between the tool dimensions and the component dimensions prior to the start of production. There shall be adequate surveillance to assure that the drawing requirements are being met throughout the life of the tool.

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Conformance Inspection by classification of characteristics

PARAGRAPH 4.3.2.1	TITLE Plug, Shock Attenuating	SHEET 1 OF 2	DRAWING NUMBER 9341742
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Width of eye section, min.	Level IV	3.2.1	AIE
102	Length of thread section	Level IV	3.2.1	AIE
103	Dimension from bottom thread section to bottom of flange	Level IV	3.2.1	AIE
104	Diameter of shank	Level IV	3.2.1	AIE
105	Dimension from bottom of gasket groove to face of flange, min	Level IV	3.2.1	AIE
106	Diameter of flange	Level IV	3.2.1	AIE
107	Diameter of gasket groove	Level IV	3.2.1	AIE
108	Depth of gasket groove, max.	Level IV	3.2.1	AIE
109	True position of gasket groove	Level IV	3.2.1	AIE
110	Perpendicularity of flange	Level IV	3.2.1	AIE
111	Thread pitch diameter, min.	Level IV	3.2.1	AIE
112	Thread major diameter, min.	Level IV	3.2.1	AIE
113	Tension test	4.4.1	3.4	4.4.1.1
114	Ductility test	4.4.1	3.5	4.4.1.2
115	Fracture test	4.4.1	3.6	4.4.1.3
116	Radius at base of shank, max.	Level IV	3.2.1	AIE

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Conformance Inspection by classification of characteristics

PARAGRAPH 4.3.2.1	TITLE Plug, Shock Attenuating	SHEET 2 OF 2	DRAWING NUMBER 9341742
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Minor				
201	Thread pitch diameter, max.	Level II	3.2.1	AIE
202	Width of gasket groove	Level II	3.2.1	AIE
203	True position of flange	Level II	3.2.1	AIE
204	Radius at top of shank, min.	Level II	3.2.1	AIE
205	Height of eye hole	Level II	3.2.1	AIE
206	Width of eye hole	Level II	3.2.1	AIE
207	Angle of groove inside diameter	Level II	3.2.1	AIE
208	Diameter of tool point, max.	Level II	3.2.1	AIE
209	Radii or chamfers missing or incorrect	Level II	3.2.1	Visual
210	Paint missing, incomplete or on undersurface	Level II	3.2.1	Visual
211	Anti-sieze compound missing or incomplete, as applicable	Level II	3.2.1	Visual
212	Surface finish incorrect	Level II	3.2.1	Visual
213	Evidence of poor workmanship	Level II	3.7	Visual

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Conformance Inspection by classification of characteristics

PARAGRAPH 4.3.2.2	TITLE Plug, Lifting, for M107 and M804	SHEET 1 OF 2	DRAWING NUMBER 13015329
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Width of eye section, min.	Level IV	3.2.2	AIE
102	Length of thread section	Level IV	3.2.2	AIE
103	Dimension from bottom thread section to bottom of flange	Level IV	3.2.2	AIE
104	Diameter of shank	Level IV	3.2.2	AIE
105	Dimension from bottom of gasket groove to face of flange, min	Level IV	3.2.2	AIE
106	Diameter of flange	Level IV	3.2.2	AIE
107	Diameter of gasket groove	Level IV	3.2.2	AIE
108	Depth of gasket groove, max.	Level IV	3.2.2	AIE
109	True position of gasket groove	Level IV	3.2.2	AIE
110	Perpendicularity of flange	Level IV	3.2.2	AIE
111	Thread pitch diameter, min.	Level IV	3.2.2	AIE
112	Thread major diameter, min.	Level IV	3.2.2	AIE
113	Tension test	4.4.1	3.4	4.4.1.1
114	Ductility test	4.4.1	3.5	4.4.1.2
115	Fracture test	4.4.1	3.6	4.4.1.3
116	Plug not appropriate color	Level IV	3.2.2	AIE

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Conformance Inspection by classification of characteristics

PARAGRAPH 4.3.2.2	TITLE Plug, Lifting, for M107 and M804	SHEET 2 OF 2	DRAWING NUMBER 13015329
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Minor				
201	Thread pitch diameter, max.	Level II	3.2.2	AIE
202	Width of gasket groove	Level II	3.2.2	AIE
203	True position of flange	Level II	3.2.2	AIE
204	Radius at top of shank, min.	Level II	3.2.2	AIE
205	Height of eye hole	Level II	3.2.2	AIE
206	Width of eye hole	Level II	3.2.2	AIE
207	Angle of groove inside diameter	Level II	3.2.2	AIE
208	Diameter of tool point, max.	Level II	3.2.2	AIE
209	Radii or chamfers missing or incorrect	Level II	3.2.2	Visual
210	Paint missing, incomplete or on undersurface	Level II	3.2.2	Visual
211	Anti-sieze compound missing or incomplete, as applicable	Level II	3.2.2	Visual
212	Surface finish incorrect	Level II	3.2.2	Visual
213	Evidence of poor workmanship	Level II	3.7	Visual

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4.4 Testing.

4.4.1 Mechanical properties tests. The minimum sample size for these tests shall be 20 parts per lot. If a multiple cavity mold is used, the sample shall be divided equally among the cavities. If more than 20 cavities are used, the sample shall consist of one part from each cavity. If any sample part fails to meet any requirement, the lot represented by the failing part shall be rejected.

4.4.1.1 Tension. The lifting plug shall be screwed into a threaded plate or other suitable fixture and a load of 8,000 pounds applied through the medium of a ¾ inch diameter bolt or pin passing through the eye of the plug. The full load shall be transmitted through the screw threads of the plug.

4.4.1.2 Ductility. The plugs used in the tension test shall be cooled to minus 40 degrees Fahrenheit (F.) for a minimum of 30 minutes. The eye of each plug shall be flattened by either pressure or blows in any direction until the inner diameter of the eye is permanently reduced by 0.050 inch while at minus 40 degrees F. Each plug shall be visually examined for cracks after flattening.

4.4.1.3 Fracture. The body of each sample (the samples from 4.4.1.2 may be used, once brought back to room temperature) iron casting shall be broken through the neck section for inspection. The plug shall be observed for discontinuous structure, porosity or shrinks.

4.4.2 Salt spray. The sampling plan for this test shall be as specified in TT-C-490. The test shall be performed on the plugs as specified in TT-C-490. The number of hours and percentage of sodium chloride solution shall be as indicated in dwg. 9341742 or 13015329. Examination shall be made for any evidence of corrosion of the painted areas. If any plug fails to comply with the specified requirements, it shall be classified defective and corrective action taken as specified in TT-C-490.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DOD personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department of Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The component covered by this specification is intended for use in lifting and handling artillery projectiles. This specification contains requirements suitable for production. This specification is military unique. The shock attenuating lifting plug can be used on any 155MM HE projectile. It is made out of malleable iron. The lifting plug for M107 and M804 is to be used only on the M107 and M804 155MM projectiles because that plug does not provide the required protection other projectiles require.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. For acquisition purposes, the contract or order should specify packaging requirements. The lifting plugs should be palletized and marked in accordance with ASTM-D-3951 (Standard Practice for Commercial Packaging) and be ordered by the applicable stock number.

c. Requirements for submission of first article sample.

d. Submission of contractor acceptance inspection equipment (AIE)(see 6.3).

e. Establishment of visual defect standards (see 6.5).

f. Submission of ammunition data cards (see 6.6).

6.3 Submission of contractor acceptance inspection equipment (AIE) designs for approval. Submit copies of designs as required to: Commander, U.S. Army TACOM-ARDEC, ATTN: AMSRD-AAR-QEM-A, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.4 Visual standards. Visual standards for evaluation of metal defects will be established for each contract after the first month of production.

6.5 Submission ammunition data cards. In addition to the normal distribution of records for items procured by the Department of the Army, one copy of all ammunition data cards should be forwarded to Commander, TACOM-ARDEC, ATTN: AMSRD-AAR-QEM-A, and AMSRD-AAR-AEM-C, Picatinny Arsenal, NJ 07806-5000 and Army Master Depository, Rock Island, IL 61299-6000.

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6.6 Inspection lots. It is noted that the size of inspection lots of components, assemblies or items of delivery may differ from the actual quantities contractually scheduled for delivery. However, in order to facilitate scheduling of tests or deliveries, inspection lots of items of delivery may be equivalent to such contract quantities provided all of the lot formation criteria per MIL-STD-1168 and sampling provisions of this specification are maintained.

6.7 Visual examination qualification. When compliance with the applicable requirement is in doubt as a result of visual examination the characteristic may be measured or gaged to determine acceptability.

6.8 Alternative testing procedures. Alternative testing procedures may be used by the contractor when such testing and procedures provide as a minimum the quality assurance provisions specified herein. Prior to applying the alternative procedures, the contractor should describe them in a written proposal and should demonstrate for the approval of the responsible Government technical agency that their effectiveness is equal to, or better that the quality assurance provisions specified herein.

6.9 Dimensional control of cast or automatic tool parts. In place of the normal sampling associated with the classification of defects, and subject to the approval of the government procuring agency, the optional sampling plans in paragraphs 6.10.1 and 6.10.2 may be used for dimensional control of cast or automatic tool parts. Use of these optional plans is limited to those dimensions and features controlled by the tool, die or mold cavity. For all specified test, visual examinations, and examinations of dimensions or features not developed exclusively by the tool, die or mold cavity, these optional plans are not applicable. In the event of in-process reject(s) during optional sampling, corrective action applicable to parts produced prior to the rejected, and subsequent to the last acceptable sample, should require approval of the government procuring agency.

6.9.1 Cast parts. A sample of at least five parts (as cast) from each cavity should be dimensionally inspected to qualify a new or reworked cavity for use in production or as a control of the cavity during production. Inspection for control of each cavity during production should be performed on the above quantities from each cavity after the production of 5000 parts. Individual cavity identification must be provided. If any defective parts are found during the qualification of the cavity, the cavity producing the defective parts should not be used in production. If any defective parts are found when inspection is performed for control of the cavity, the cavity producing the defective part should be removed from production. Further, that portion of production since the last control check should be returned for corrective action. All cavities formerly removed from production because of some fault, may, after reworking, be

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returned to production providing they pass the qualification test specified above.

6.9.2 Automatic tool parts (see 6.11.b). A sample of at least 10 parts should be dimensionally inspected to qualify a new or adjusted setup for use in production. For multiple simultaneous operations (the same operation on 2 or more parts), at least 5 parts per operation per setup should be inspected. In addition, a random sample of 5 parts per setup or simultaneous operation should be selected from the last portion of each hour's production, for dimensional inspection, as a control of the setup during production. If an hour's production is 250 parts or less per setup or simultaneous operation, the hourly sample may be reduced to 3 parts per setup or simultaneous operation. Automatic tools should be considered for this reduced inspection only when equipped with devices to provide automatic shut-down or clear immediate warning if machine power input (electricity, air or oil pressure) fall below the minimum level necessary for normal machine operation. All parts in process when power input runs below minimum should be inspected for dimensional requirements prior to acceptance. Such parts should not be counted as qualification or control samples. If any defective parts are found during qualification of a setup, that setup should not be used in production. If any defective parts are found when inspection is performed for control of a setup, the setup producing the defective part should be removed from production. Further, that portion of production since the last setup control check should be returned for corrective action. All setups removed from production because of some fault may, after adjustment or re-tooling as necessary, be returned to production providing they pass the qualification test above.

6.10 Definitions

a. Acceptance Inspection Equipment (AIE). Any measuring device that is traceable to the national or international standard which is used to assure conformance of material to the contract requirements.

b. Automatic tool. An automatic tool is any self-regulating piece or assemblage of production equipment which is setup or programmed to perform one or more operation on a part. The tool operator's functions should not exceed part handling and input of a start signal. The tool setup should provide positive, repetitive, three plane positioning of the work piece.

6.11 Changes from previous issues. Marginal notations are not used in this version to identify changes with respect to the previous issue due to the extent of the change

6.12 Subject term (keyword) listing.

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Salt Spray
Malleable Iron
Ductile Iron
Ductility
Blue Lifting Plug

Custodian:
Army – AR

Preparing Activity:
Army – AR
(Project 1320-2007-003)

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